A RECURSIVE BUCHSTAB SWITCHING IMPROVEMENT FOR THE GREATEST PRIME FACTOR OF $n^2 + 1$

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Abstract. We recursively expose the one-prime Buchstab tail on $7/6 < \alpha < 5/4$, treating its children with the Grimmelt–Merikoski Type I/II estimates and the dimension-one linear sieve. The resulting switching saves at least 0.032303187971 on the normalized sieve scale. Granting these analytic inputs and the finite Mellin–Perron transfer, infinitely many n satisfy $P^+(n^2+1) > n^{1.323}$. This is a greatest-prime-factor weak form of Landau's fourth problem; the parity barrier remains.

1. Introduction

Landau's fourth problem asks whether $n^2 + 1$ is prime for infinitely many positive integers n. Iwaniec proved that $n^2 + 1$ is a prime or a product of two primes for infinitely many n [4]; the possibility that all surviving values are semiprimes is the classical parity obstruction. A standard quantitative complement is to make $P^+(n^2 + 1)$ as large as possible. Merikoski proved that $P^+(n^2 + 1) > n^{1.279}$ infinitely often and obtained 1.312 conditionally on Selberg's eigenvalue conjecture [6]. Pascadi obtained the unconditional exponent 1.3 [7]. Grimmelt and Merikoski then recovered 1.312 unconditionally by proving uniform Type I and Type II estimates for roots of quadratic congruences [2]. Li announced 1.317, using sharper numerical bounds for the same Buchstab decomposition [5].

These exponents quantify progress toward Landau's problem: a bound $P^+(n^2 + 1) > n^{\vartheta}$ leaves a cofactor $m < n^{2-\vartheta+o(1)}$. Theorem 1.1 lowers this exponent to 0.677, while still leaving the parity distinction $m = 1$ versus $m > 1$ unresolved.

Our new input is a recursive Buchstab switching in the range where the old argument used the linear sieve. Merikoski already observed that Buchstab's identity could be applied more times there to generate additional Type II sums [6, Remark 12]. With the stronger range now available unconditionally, this produces a numerically substantial saving. In addition to retaining Type II three-prime children, we use the lower linear sieve for the complementary children and recursively upper-bound the two-prime descendants of the previously discarded one-prime tail. Li explicitly keeps the same sieve decomposition and changes only the numerical estimates for its terms [5, Lemmas 2.1 and 2.2]; in particular, the contribution on $7/6 < \alpha < 5/4$ remains the old linear-sieve term minus the one-prime Type II term. The three-prime term below is therefore absent from the previously announced 1.317 argument.

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Theorem 1.1. *Assume the analytic inputs in Sections 2 and 3. Specifically, assume the Type I/II estimates of Grimmelt and Merikoski, the dimension-one linear sieve, and the finite Mellin–Perron transfer. Then there are infinitely many positive integers n such that*

$$P^+(n^2+1) > n^{1323/1000} = n^{1.323}.$$

Remark 1.2 (Proof-status boundary)*.* The hypotheses are the analytic inputs named in the theorem; they are not formalized in Lean. Lean checks only finite algebra, proof-adverse cell geometry, the integer certificate, endpoint arithmetic, and the real-power conversion. It neither verifies those analytic inputs nor solves Landau's fourth problem.

Remark 1.3 (Comparison with an isolated larger claim)*.* Carella's arXiv note [1, Section 5, (5.3)–(5.4)] claims the exponent 3/2, but the transition from its (5.3) to (5.4) replaces

$$\sum_{d|m} \Lambda(d) = \log m \quad \text{by} \quad \sum_{p|m} \log p = \log \operatorname{rad}(m)$$

while retaining a lower bound. The latter quantity is at most the former, and n^2+1 need not be squarefree (for example, $7^2+1 = 2\cdot 5^2$). That step therefore does not establish the claimed bound. Our comparison frontier is the Grimmelt–Merikoski theorem and Li's subsequent 1.317 calculation.

The gain over 1.317 is 0.006. The argument is computer assisted only in the evaluation of an explicit finite family of piecewise-rational integrals. The certificate uses no floating-point operations. We retain Merikoski's original, relatively loose upper bounds for all positive Buchstab deficiencies before 7/6, so Li's Mathematica integrations are not an input.

2. Transferred arithmetic information

We use the notation of [6, Section 2]. Let b be a fixed nonnegative smooth function supported on $[x, 2x]$, put

$$|\mathcal{A}_d| = \sum_{\ell^2+1\equiv 0 \pmod d} b(\ell), \qquad X_b = \int_{\mathbb{R}} b(t)\,dt,$$

and let

$$\rho(d) = \#\{\nu \pmod d : \nu^2+1 \equiv 0 \pmod d\}.$$

The model sequence replaces $|\mathcal{A}_d|$ by $X_b\rho(d)/d$. For a smooth modulus weight ψ_P supported on $[P, 4P]$, write

$$S(x,P) = \sum_p \psi_P(p)|\mathcal{A}_p|\log p.$$

Proposition 2.1 (Grimmelt–Merikoski transfer)**.** *Fix a sufficiently small interior margin $\eta > 0$. The Type I estimate used in* [6, Proposition 3] *is available with $D \le x^{1/2-\eta}$. If $MN = x^\alpha$ and $M \ge N$, the Type II estimate in* [6, Proposition 4(i)] *is available, with a power-saving error, throughout every closed interior subrange of*

$$x^{\alpha-1} < N < x^{(2-\alpha)/3}, \tag{1}$$

for divisor-bounded coefficients whose N-coefficient is supported on squarefree integers.

Proof. Take $a = h = 1$ in Grimmelt–Merikoski. The paragraph immediately following their Theorem 1.1 on PDF p. 2 says that, when $ah \le x^{\varepsilon}$, its auxiliary prime-density hypothesis follows unconditionally from the classical zero-free region (a possible exceptional zero only decreases the relevant sum). Thus no character hypothesis remains for the present fixed shift.

Corollary 7.1 on PDF p. 22 is stated for $D \le x^{1/2-\eta}$ and gives the sum over $d \le D$ of the absolute value of the complete smooth modulus-d discrepancy. Its test functions are normalized on $[1, 2]$ and $[-1, 1]$, with constants controlled by finitely many derivative seminorms. Every dyadic piece used here is obtained by rescaling one fixed smooth partition, so those seminorms, and hence the implied constant, are uniform in the piece. The $O((\log x)^L)$ pieces and Mellin kernels below cost only a power of $\log x$.

Corollary 7.2 on PDF pp. 22–23 is stated for divisor-bounded coefficients, with the N coefficient squarefree-supported, and gives

$$x^{\alpha-1+2\eta} \le N \le x^{(2-\alpha)/3-4\eta/3}.$$

In every application below, N is the product of a selected subset of at most three distinct prefix primes. Hence it is squarefree and has at most $\tau_3(N)$ ordered representations. The complementary primes, the Möbius variable, and the residual cofactor give an M coefficient bounded by $\tau_5(M)$. The localization factors have bounded real part and unit modulus on vertical lines, so these same fixed divisor bounds survive the finite Mellin–Perron separation. Thus the localized sums satisfy the coefficient hypotheses of Corollary 7.2 term by term. Also $N \le x^{1/3}$, so $M = x^{\alpha}/N \ge N$ in all ranges used here. Since η is fixed and positive, the displayed shifted interval is a closed interior subrange of (1); conversely every fixed closed interior subrange is obtained by taking η small enough.

More explicitly, the error exponents in Corollaries 7.1 and 7.2 are, up to an $x^{o(1)}$ factor,

$$1 - (1 - 2\theta)\eta + \varepsilon/4, \qquad 1 - (1 - 2\theta)\eta + \varepsilon/8.$$

Unconditionally $\theta = 7/64$. After fixing the sieve margin, choosing the auxiliary $\varepsilon = \eta$ gives respectively $1 - 17\eta/32$ and $1 - 21\eta/32$; the logarithmic losses below are therefore absorbed by a fixed power saving. The paragraph after Corollary 7.2 on PDF p. 23 explicitly says that these are the Type I/II ranges used in the proof of [6, Theorem 2]. This checks all hypotheses needed from the two corollaries; their proofs remain the cited external results. □

We also use Merikoski's Fundamental Proposition II with Selberg parameter zero. In the present notation it evaluates

$$\sum_{u \le x^{\xi-\eta}} \lambda_u S(\mathcal{A}(P)_u, x^{\gamma-\eta}) \tag{2}$$

by its model, for divisor-bounded λ_u, where

$$\sigma = \frac{2-\alpha}{3}, \qquad \gamma = \sigma - \alpha + 1 = \frac{5-4\alpha}{3}, \qquad \xi = \frac{3}{2} - \alpha. \tag{3}$$

This follows from Proposition 2.1 by the proof of [6, Proposition 6].

3. The recursive switching

Fix $7/6 < \alpha < 5/4$ and put

$$a = \alpha - 1, \qquad \tau = \frac{\xi}{2}.$$

Then $0 < \gamma < \tau < a < \sigma$. We suppress the harmless interior margins in the displayed limiting formulae, but record one compatible choice here. If $\nu > 0$ is the parameter in Corollary 7.2 of [2], put

$$\begin{aligned} &a_\nu = a + 3\nu, \quad \sigma_\nu = \sigma - 2\nu, \\ &\gamma_\nu = \gamma - 4\nu, \quad \xi_\nu = \xi - 3\nu, \qquad \tau_\nu = \xi_\nu/2. \end{aligned} \tag{4}$$

The available Type II interval is $[a + 2\nu, \sigma - 4\nu/3]$, so $[a_\nu, \sigma_\nu]$ lies strictly inside it. The proof of the fundamental proposition works uniformly for $u \le x^{\xi_\nu}$ and sieve level x^{γ_ν}: its Type I part has

$$x^{a+2\nu} x^{\xi_\nu} = x^{1/2-\nu},$$

while the first Type II prefix in its complementary part is smaller than

$$x^{a+2\nu+\gamma_\nu} = x^{\sigma-2\nu} < x^{\sigma-4\nu/3}.$$

Thus the two published corollaries give exactly this margin version, not merely the limiting endpoint. We use the fixed value $\nu = \nu_0 = 10^{-20}$ from Lemma 4.3. On the compact range $\alpha < 1.22$ used by the certificate, $4\nu_0 < 1/25 \le \gamma$ and $5\nu_0 < 1/25 \le \sigma - a$, so $0 < \gamma_{\nu_0} < \tau_{\nu_0} < a < \sigma$. All analytic estimates are taken with this fixed positive margin; no limiting choice of ν is used in the endpoint argument.

3.1. **Uniform localization and prefix sieving.** We record two standard uniformity facts because the recursive decomposition uses them in combinations not written explicitly in the earlier papers. The first is the finite Mellin–Perron device used below.

Lemma 3.1 (Finite cross-condition localization)**.** *Fix $R, C, J \ge 1$. Consider a dyadically localized bilinear sum with $MN \asymp x^\alpha$, whose two coefficients are bounded by τ_R, and impose at most J strict conditions*

$$U(\boldsymbol{m}) < V(\boldsymbol{n}).$$

Here U and V are positive integer-valued functions of the variables assigned to the M- and N-coefficients, respectively, and $U, V \le x^C$. A fixed smooth function of MN/P and a factor $\log(MN)$ may also be present. Then the sum is a multiple Mellin–Perron integral of ordinary Type II sums whose coefficients are bounded by $O_{C,J}(\tau_R)$. The total L^1 mass of the kernels is $O((\log x)^{O_{C,J}(1)})$, and the truncation error is $O_A(x^{-A})$ for any fixed $A > 0$, after increasing the truncation height. Consequently any fixed power-saving Type II estimate that is uniform for divisor-bounded coefficients survives all these localizations.

Proof. Since U, V are integers, the strict comparison is equivalent to $U < V - 1/2$. Put $y = (V - 1/2)/U$ and $c = 1/\log x$. For $y \ne 1$, the truncated Perron formula is

$$\frac{1}{2\pi i} \int_{c-iT}^{c+iT} y^s \frac{ds}{s} = \mathbf{1}_{y>1} + O\left(y^c \min\left\{ 1, \frac{1}{T|\log y|} \right\} \right).$$

Here is the required estimate, rather than an appeal to an unquantified localization principle. The full vertical integral equals 1 for $y > 1$ and 0 for $0 < y < 1$: close the contour to the left or right, respectively, and take the residue at $s = 0$. On either omitted vertical tail, integration by parts in $e^{it \log y}/(c + it)$ gives $O(y^c/(T|\log y|))$ when $T|\log y| \geq 1$; the trivial bound for the complete transition gives $O(y^c)$ otherwise. This proves the displayed formula.

The half-integer cut makes the estimate uniform. If $U < V$, then $y \geq 1 + 1/(2U)$ and

$$\log y \geq \frac{2}{4U + 1}.$$

If $U \geq V$, then $1 - y \geq 1/(2U)$ and $-\log y \geq 1 - y \geq 1/(2U)$. Hence in both cases

$$|\log y| \geq \frac{2}{4U + 1} ge \frac{2}{5x^C}.$$

Moreover $y^c \leq e^C$ when $y > 1$, and $y^c \leq 1$ when $y < 1$. Thus one comparison has pointwise truncation error $O_C(x^C/T)$.

The replacement of V by $V - 1/2$ only excludes the equality case. It does not change the underlying representations or their divisor bounds.

For a cross-condition $U(\boldsymbol{m}) < V(\boldsymbol{n})$, attach U^{-s} to the M-coefficient and $(V - 1/2)^s$ to the N-coefficient. If the smaller variable belongs to the N side, attach U^{-s} there and $(V - 1/2)^s$ to M; the designation of the squarefree Type-II family is not changed. A product cutoff has the same form: if $A_M A_N \leq Q$, with A_M and A_N formed from variables on the two sides, write it as

$$A_M < \lfloor Q/A_N \rfloor + 1.$$

Thus $U = A_M$ is assigned to the M side and $V = \lfloor Q/A_N \rfloor + 1$ to the N side; the complementary inequality is handled by subtracting this class. This also covers any exact priority cutoff not already fixed by the exponent cell. If U or V depends on a particular prime-factor representation rather than only on the collected integer, attach the phase before collecting that coefficient. The number of prime slots is fixed and the vertical factors are uniformly bounded, so summing those representations preserves a fixed divisor bound.

The recursive branches specialize this rule as follows.

- **A_3, Type II.:** For the first eligible subset I, put $N = \prod_{i \in I} q_i$ and $M = (\prod_{i \notin I} q_i) ds$. An ordering comparison split across the families assigns the smaller prime to U^{-s} and the larger to $(V - 1/2)^s$. If $q_3 \in N$, roughness is $P^+(d) < q_3$, with $P^+(d)^{-s}$ on M and $(q_3 - 1/2)^s$ on N; if $q_3 \notin N$, it is internal to M. Priority product cutoffs are fixed away from cell boundaries, or are separated by the preceding $A_M A_N \leq Q$ rule.
- **Direct pair Type II.:** Take $N = q_1 q_2$ and put the sifted cofactor and Mobius variable on M. The order $q_2 < q_1$ is internal to N; for $d > 1$, the only cross condition is $P^+(d) < q_2$, assigned as above. The term $d = 1$ is automatic.
- **Type-I prefix branches.:** The A_3 lower-sieve branch has prefix $q_1 q_2 q_3$ and cutoff q_3; the universal U_{LS} branch has prefix $q_1 q_2$ and cutoff q_2. Their ordering is internal and the variable cutoff is carried by the prefix-dependent Rosser weight of Proposition 3.2; neither branch invokes this Type-II Perron lemma.

Fixed-cutoff bases.: The one-prime tail base has prefix q_1, and the recursive base has prefix q_1q_2, both with cutoff x^{γ_ν}. Their cell conditions are fixed before the Fundamental Proposition is applied. The recursive children reuse the A_3 assignments above; the choice between direct and recursive upper bounds is a boxwise decision, not a Perron condition.

For each priority class the number J of genuine cross-comparisons is fixed, independent of x. On $\Re s = c$, the real factors are at most e^C and the imaginary factors have modulus one. Also

$$\int_{-T}^{T} \frac{dt}{|c+it|} \ll \log(T\log x) \ll_{A'} \log x \qquad (T = x^{A'}).$$

Replace the J indicators one at a time and telescope. If the unreduced sum has at most x^B tuples, the total truncation error is

$$O_{C,J}\left(x^B \frac{x^C}{T}(\log x)^{J-1}\right).$$

Given $A > 0$, choosing $A' > A + B + C + J + 2$ makes this $O_A(x^{-A})$. The resulting multiple integral has total kernel mass $O((\log x)^J)$. Therefore a Type II error $O(x^{1-\delta})$, uniform for divisor-bounded coefficients, becomes $O(x^{1-\delta}(\log x)^J) = O(x^{1-\delta/2})$ for large x.

For a smooth compactly supported W, Mellin inversion separates $W(MN/P)$; rapid decay of its Mellin transform gives the same arbitrary power truncation error. Finally $\log(MN) = \log M + \log N$, with the logarithms handled by partial summation. All vertical factors have size $O_C(1)$, so the coefficients remain uniformly divisor bounded. The fixed polylogarithmic kernel mass is absorbed after decreasing the power-saving exponent. This proves the lemma. Merikoski's use of Perron in the proof of Fundamental Proposition II and his Mellin removal of the smooth product weights provide the qualitative templates [6, Sections 2.3 and 3.4]. This proof is an ordinary analytic argument and is not part of the Lean formalization. $\square$

Proposition 3.2 (Prefix-uniform linear sieve)**.** *Let a dyadically localized ordered prefix $\boldsymbol{q} = (q_1, \ldots, q_j)$ consist of at most three primes. Put $u = q_1 \cdots q_j$ and suppose that the localized cell gives the uniform upper bound $u \le x^{s_0}$, with $1/2 - s_0 - \nu > 0$. All prefix primes are at least the sieve cutoff $z = z(\boldsymbol{q})$, with $z(\boldsymbol{q}) \asymp x^\zeta$; in the variable cutoff branches, $z(\boldsymbol{q}) = q_j$ is the least prefix prime. The Type I estimate in Proposition 2.1, summed over any fixed finite collection of such cells with divisor-bounded prefix weights, permits the standard dimension-one upper and lower linear sieve with remainder level*

$$D = x^{1/2-s_0-\nu}.$$

After removal of the common Euler factor, the corresponding sieve parameter is $s = (1/2 - s_0 - \nu)/\zeta$ and the main-term multipliers are $e^{-\gamma_E}F(s)$ and $e^{-\gamma_E}f(s)$. The lower multiplier is understood to be zero for $s \le 2$. The aggregate remainder is $O(x^{1-\delta})$ for some $\delta = \delta(\nu) > 0$.

Proof. For each localized prefix, insert the upper or lower Rosser weights at level $D = x^{1/2-s_0-\nu}$. The per-prefix inequalities are the dimension-one formulae (12.12)–(12.14) of [3, Chapter 12, pp. 235–236]. Their remainders are kept with their signs. After multiplying by the nonnegative prefix partition and summing, put $k = um$

and $e = ud$. This gives, before any absolute value is taken,

$$\sum_{\boldsymbol{q}} c(\boldsymbol{q}) R^{\pm}_{\boldsymbol{q}} = \sum_{e} \Lambda^{\pm}_{e} r_P(e).$$

Here $r_P(e)$ is exactly the smooth modulus-e discrepancy occurring inside the absolute value in Grimmelt–Merikoski Corollary 7.1; in particular it is independent of the prefix representation of e. Appendix A proves

$$|\Lambda^{\pm}_{e}| \ll (\log x)^B \tau_K(e), \qquad ud \le x^{1/2-\nu},$$

so the absolute discrepancy sum in [2, Corollary 7.1, p. 22], with $a = h = 1$, leaves a fixed power saving after all prefix cells are summed.

The appendix also evaluates the model sums uniformly. The primes in a Rosser modulus are below the least prefix prime, so they are coprime to u; Merikoski's Lemma 8 and the dimension-one fundamental lemma therefore give the same Euler product as in the unprefixed sequence. For the A_3 lower branch the three-prime level remains a positive power of x at $\alpha = 7/6$, while $s \le 2$ contributes zero. For the two-prime upper branch the resulting coefficient is U_{LS}; if $0 < s < 1$, positivity allows the cutoff to be lowered to $z_* = D^{1-\omega}$ before applying the upper sieve. Finally Merikoski's Lemma 7, with the sieve multiplier frozen on each exponent cell, converts the model sums to (12) and (13). The hypotheses retained in Remark 3.3 are precisely the cited external analytic theorems, rather than an additional cancellation assumption. □

Remark 3.3 (Limitation of the prefix-sieve transfer). Proposition 3.2 is not a Lean theorem. Its external inputs are precisely Grimmelt–Merikoski Corollary 7.1 and the dimension-one fundamental lemma of *Opera de Cribro*; neither source theorem is reproved here. Appendix A checks their hypotheses for the least-prefix cutoff, including the uniform Euler product, the normalized smooth family, and the collected absolute remainder. Thus no additional "prefix-uniformity" assertion is left as an assumption.

The decomposition preceding the old linear-sieve term is

$$S(\mathcal{A}(P), 2\sqrt{P}) \le S(\mathcal{A}(P), x^a) - \sum_{x^a < q < x^\sigma} S(\mathcal{A}(P)_q, q). \tag{5}$$

The negative sum in (5) is retained unchanged; its model contribution is the term F_6 bounded in Section 5. We improve only the first term. The first Buchstab split is

$$\begin{aligned} S(\mathcal{A}(P), x^a) &= S(\mathcal{A}(P), x^\gamma) - \sum_{x^\gamma < q_1 < x^\tau} S(\mathcal{A}(P)_{q_1}, q_1) - R_1, \\ R_1 &:= \sum_{x^\tau < q_1 < x^a} S(\mathcal{A}(P)_{q_1}, q_1). \end{aligned} \tag{6}$$

Applying Buchstab twice more to the retained sum yields

$$S(\mathcal{A}(P), x^a) = A_0 - A_1 + A_2 - A_3 - R_1, \tag{7}$$

where

$$
\begin{aligned}
A_0 &= S(\mathcal{A}(P), x^\gamma),\\
A_1 &= \sum_{x^\gamma < q_1 < x^\tau} S(\mathcal{A}(P)_{q_1}, x^\gamma),\\
A_2 &= \sum_{x^\gamma < q_2 < q_1 < x^\tau} S(\mathcal{A}(P)_{q_1 q_2}, x^\gamma),\\
A_3 &= \sum_{x^\gamma < q_3 < q_2 < q_1 < x^\tau} S(\mathcal{A}(P)_{q_1 q_2 q_3}, q_3).
\end{aligned}
$$

The first three terms are covered by (2), since $q_1 \le x^\tau$ and $q_1 q_2 \le x^{2\tau} = x^\xi$.

For A_3, retain only tuples for which at least one of

$$\beta_1 + \beta_2, \quad \beta_1 + \beta_3, \quad \beta_2 + \beta_3, \quad \beta_1 + \beta_2 + \beta_3 \tag{8}$$

lies in the shifted Type II interval $[a_\nu, \sigma_\nu]$. The corresponding product of distinct primes is squarefree and may be chosen as the Type II variable. Use the canonical $1200 \times 300 \times 160$ exponent mesh of Proposition 4.1. A cell meeting a Type II endpoint or a priority boundary is assigned the proof-adverse lower contribution (zero if no other lower sieve is available). Thus its loss is already included in the certified saving, rather than being postponed to a limiting mesh argument. On every remaining cell, Lemma 3.1 removes the finitely many ordering and roughness cross-conditions and leaves divisor-bounded coefficients. This Type II part gives one lower bound for A_3. Unlike the earlier three-step argument, we also lower-bound complementary triples by the dimension-one lower linear sieve and take the larger available weight. Since A_3 occurs with a negative sign, every such lower bound may be subtracted in (7).

Lemma 3.4. *After restoring fixed interior margins, the retained part of A_3 has the same asymptotic formula for $\mathcal{A}(P)$ and $\mathcal{B}(P)$, with a power-saving error on each smooth P-block.*

Proof. Order the four subsets in (8) and assign each good prime triple to the first subset whose exponent sum lies in the fixed interior of $[a_\nu, \sigma_\nu]$. For every earlier subset, nonmembership is the disjoint union of the alternatives "below the lower endpoint" and "above the upper endpoint". Expanding these alternatives produces only finitely many priority classes (at most $1 + 2 + 2^2 + 2^3$ before empty classes are removed). Smoothly localize the three prime variables. For an assigned subset I, put $N = \prod_{i \in I} q_i$ and put the complementary prime factors and the sifted cofactor into M. Then $MN \asymp x^\alpha$,

$$x^{a_\nu} \le N \le x^{\sigma_\nu}, \qquad M \ge N,$$

and N is squarefree. The coefficient of N has only divisor-boundedly many prime representations. The coefficient of M is also divisor bounded, since it records at most the complementary prime factors and one cofactor.

Here are the coefficient and roughness details. From the definition of a sifted sum,

$$S(\mathcal{A}(P)_{q_1 q_2 q_3}, q_3) = \sum_{(r, P(q_3)) = 1} |\mathcal{A}_{q_1 q_2 q_3 r}| \psi_P(q_1 q_2 q_3 r) \log(q_1 q_2 q_3 r).$$

Insert the finite identity

$$\mathbf{1}_{(r,P(q_3))=1} = \sum_{d|r,\, d|P(q_3)} \mu(d)$$

and write $r = ds$. If $N = \prod_{i\in I} q_i$ is the selected subset, put $M = (\prod_{i\notin I} q_i)ds$. After dyadic subdivision, the resulting sum has exactly the bilinear form of Proposition 2.1. The N-coefficient is supported on squarefree integers: the selected q_i are distinct. More quantitatively, if $t = |I|$, then before the separation of cross-conditions the numbers of possible factorizations obey the uniform bounds

$$\#\{N = \textstyle\prod_{i\in I} q_i\} \le \tau_t(N), \qquad \textstyle\sum_{M=(\prod_{i\notin I} q_i)ds} |\mu(d)| \le \tau_{5-t}(M).$$

The harmless conventions $\tau_0(1) = 1$ and $\tau_j \le \tau_5$ may be used. Thus both coefficients are bounded by a fixed divisor function. Repeated prime factors shared by M and N do not affect this conclusion: Corollary 7.2 requires squarefree support only for the N coefficient and imposes no coprimality condition on M, N.

There is one genuine cross-condition when q_3 belongs to the selected product N. For $d > 1$ the condition $d \mid P(q_3)$ is equivalently

$$\mu(d) \ne 0, \qquad P^+(d) < q_3;$$

the term $d = 1$ is automatic. Hence the only datum from the M side which must be compared with the N side is the single integer $P^+(d)$. A truncated Perron integral separates $P^+(d) < q_3$ into $(P^+(d))^{it}$ and q_3^{-it}. If $q_3 \notin N$, this condition is already internal to the M coefficient. This makes explicit that expanding the roughness condition does not create an unbounded number of cross variables.

It remains to separate conditions that involve variables on both sides. Prime ordering and the strict comparison $P^+(d) < q_3$ have exactly the integer form covered by Lemma 3.1; using $q_3 - 1/2$ in the latter comparison also removes any equality ambiguity in truncated Perron inversion. The selected-subset priority is already fixed on the interior exponent cell. The product cutoff and logarithm are handled by the Mellin and partial-summation separations in that lemma. Hence the two coefficient bounds above remain $O(\tau_5(M))$ and $O(\tau_5(N))$, while all kernel and dyadic losses are powers of $\log x$. These are absorbed by the fixed power saving in Proposition 2.1.

This is precisely the mechanism in the proof of Merikoski's Fundamental Proposition and in the evaluation of its three-prime term [6, Sections 2.3, 2.4.3, and 3.4]. Most importantly, the paper explicitly states there that *any combination* of q_1, q_2, q_3 lying in the Type II range may be used. The new term changes only the outer exponent boxes; every retained box lies in the unconditional interior supplied by Proposition 2.1. Summing the four disjoint priority classes proves the lemma. □

All conversions of model almost-prime sums below use exactly [6, Lemma 7]: its rough-number input is equation (2.4), its measure is $d\beta_1 \cdots d\beta_k/(\beta_1 \cdots \beta_{k-1}\beta_k^2)$, and the numerical Buchstab bounds used here are precisely equation (2.5). No stronger model-integral statement is assumed.

Let ω be the Buchstab function and set

$$u_+ = 0.5617, \qquad u_- = 0.5607.$$

The bounds

$$\omega(v) \le u_+ \quad (v \ge 4), \qquad \omega(v) \ge u_- \quad (v \ge 2.47) \tag{9}$$

are quoted in [6, (2.5)]. On the retained range $\alpha < 1.22$, an ordered three-prime child has
$$\frac{\alpha-\beta_1-\beta_2-\beta_3}{\beta_3} > \frac{3-2\alpha}{\alpha-1} > 2.54.$$
The one- and two-prime fundamental-proposition bases have arguments at least 9 and 7.5, respectively. If $\beta_1+\beta_2 \le \sigma$, the ordered Type II pair has argument at least $2(\alpha-\sigma)/\sigma \ge 6.4$. Thus every use of (9) is within its stated range.

We also use the standard dimension-one linear-sieve functions F and f [3, Chapter 12]. Define the rational lower envelope
$$\Phi(s) = \begin{cases} 0, & s < 2, \\ 4(s-2)/s^2, & 2 \le s \le 4, \\ 1/2, & s \ge 4. \end{cases} \tag{10}$$
For $2 \le s \le 4$ the identity $e^{-\gamma_E} f(s) = 2\log(s-1)/s$ and $\log(1+t) \ge 2t/(2+t)$ give
$$\Phi(s) \le e^{-\gamma_E} f(s). \tag{11}$$
At $s = 4$ the rational value is $1/2$, and the standard monotonicity of f gives the continuation in (10). Similarly, $e^{-\gamma_E}F(s) = 2/s$ for $1 \le s \le 3$ and is at most $2/3$ for $s \ge 3$.

For an ordered triple put
$$s_3 = \frac{1/2-\beta_1-\beta_2-\beta_3}{\beta_3}, \qquad W_3 = \max\{u_- \mathbf{1}_{\text{good}}, \Phi(s_3)\},$$
where $\mathbf{1}_{\text{good}}$ indicates that one of (8) lies in $[a,\sigma]$. Proposition 2.1 handles the first weight; Proposition 3.2 handles the second. Indeed, after a fixed prefix of total exponent s_0, the available sieve level is $x^{1/2-s_0-\nu}$, and combining its Rosser weight with at most three prime indicators still gives a fixed divisor-bounded coefficient. Hence
$$A_3^{\text{low}}(\alpha) = \alpha \int_{\gamma<\beta_3<\beta_2<\beta_1<\tau} W_3 \, \frac{d\beta_1\, d\beta_2\, d\beta_3}{\beta_1\beta_2\beta_3^2} \tag{12}$$
is a valid limiting lower model for A_3.

It remains to retain part of R_1. For $\tau < \beta_1 < a$, Buchstab gives
$$S(\mathcal{A}(P)_{q_1}, q_1) = S(\mathcal{A}(P)_{q_1}, x^\gamma) - \sum_{x^\gamma < q_2 < q_1} S(\mathcal{A}(P)_{q_1q_2}, q_2).$$
Put $\delta_2 = 1/2-\beta_1-\beta_2$ and define the proof-adverse upper linear-sieve envelope
$$U_{\text{LS}}(\beta_1,\beta_2) = \max\left\{\frac{2}{\delta_2}, \frac{2}{3\beta_2}\right\}.$$
Let $U_2(\beta_1,\beta_2)$ be the minimum of U_{LS} and the following available upper bounds:
$$\frac{u_+}{\beta_2}, \qquad \text{if } \beta_1+\beta_2 \in [a,\sigma],$$
$$\frac{u_+}{\gamma} - \int_\gamma^{\beta_2} W_3(\beta_1,\beta_2,\beta_3)\frac{d\beta_3}{\beta_3^2}, \qquad \text{if } \beta_1+\beta_2 \le \xi,$$
where the second candidate is used only when it is nonnegative. The second line is one more Buchstab identity: Fundamental Proposition II evaluates its base and

(12) lower-bounds its children. The proof-adverse linear-sieve option is uniform over the two-prime prefix by Proposition 3.2. Consequently

$$T(\alpha) = \alpha \int_{\tau}^{a} \left[\frac{u_-}{\gamma} - \int_{\gamma}^{\beta_1} U_2(\beta_1, \beta_2) \frac{d\beta_2}{\beta_2} \right]_+ \frac{d\beta_1}{\beta_1} \tag{13}$$

is a lower model for R_1.

3.2. **Complete branch-to-input ledger.** Put

$$\mathfrak{M}(P) := X_b \int_0^{\infty} \psi_P(u) \, \frac{du}{u}.$$

This is the common smooth block factor in [6, Lemma 7]. For a fixed $\nu > 0$ and a fixed exponent mesh, let $H_\nu(\alpha)$ denote the expression below with the shifted parameters in (4) and the proof-adverse cell endpoints used by the certificate.

Proposition 3.5 (Analytic branch ledger)**.** *Uniformly for $7/6 < \alpha < 1.22$, the recursive decomposition uses exactly the one-sided estimates in the following table. Each row is valid with a power-saving remainder before the finite sums of dyadic and Perron pieces.*

term	*analytic input*	*proof direction and condition*
A_0	*Fundamental Proposition II*	*upper bound, $u = 1$, using u_+/γ*
A_1	*Fundamental Proposition II*	*lower bound on the common one-prime domain, using u_-/γ*
A_2	*Fundamental Proposition II*	*upper bound on a containing ordered two-prime domain, using $u_+/(2\gamma)$ after symmetry*
A_3	*Corollary 7.2*	*lower bound whenever a complete subset-product cell lies in $[a_\nu, \sigma_\nu]$*
A_3	*Proposition 3.2*	*lower linear-sieve bound on the same child; take the larger of the two available lower bounds*
R_1 *base*	*Fundamental Proposition II*	*lower bound because $\beta_1 < a < \xi$*
two-prime child	*Proposition 3.2*	*universal upper linear-sieve envelope U_{LS}*
two-prime child	*Corollary 7.2 or one more Buchstab split*	*take the smaller upper bound: direct Type II when $\beta_1 + \beta_2 \in [a_\nu, \sigma_\nu]$; otherwise base upper minus A_3 lower when $\beta_1 + \beta_2 \le \xi_\nu$*

Consequently, in the normalization above,

$$S(\mathcal{A}(P), x^a) \le \big(H_\nu(\alpha) + o(1)\big)\mathfrak{M}(P),$$

where the $o(1)$ is uniform on the fixed shifted cells. The quantitative comparison between this fixed-margin envelope and $H(\alpha)$ is given after Proposition 4.1; no unquantified mesh limit is used in the endpoint budget.

Proof. The signs $A_0 - A_1 + A_2 - A_3 - R_1$ determine the required directions. The first three rows follow from (2), since their prefix exponents are at most $0, \tau$, and $2\tau = \xi$, respectively. The two A_3 rows are Lemma 3.4 and Proposition 3.2; the maximum of two lower bounds for the same nonnegative child remains a lower bound.

For R_1, Buchstab's identity writes each node as a base minus its two-prime children. A base lower bound minus children upper bounds is therefore a node lower bound. Each child has the universal upper linear-sieve bound. On a complete Type II pair cell, Corollary 7.2 supplies the second upper bound. If the pair exponent is at most ξ_ν, one more Buchstab identity instead gives a base upper bound minus the already proved three-prime lower bound. Taking the minimum of valid upper bounds is valid. Lemma 3.1 preserves all these estimates after the finitely many cross-condition separations. Summing the rows proves the claim. □

Remark 3.6 (Exhaustive branch coverage). The supplementary branch-coverage audit checks the complete canonical grid using integer comparisons only. It visits 34 215 168 ordered A_3 boxes and 19 635 200 two-prime tail boxes. Every row of the ledger occurs. The proof-adverse minimum Buchstab arguments are, respectively,

$$5.206977117, \qquad 9.007506255, \qquad 8.002880298, \qquad 7.508771632$$

for the A_3 Type II lower branch, the R_1 base, the direct pair Type II upper branch, and the recursive-base upper branch. Thus the first two exceed 2.47 and the last two exceed 4, as required for the constants u_- and u_+. The audit terminates with `BRANCH_LEDGER_CERTIFIED=YES`.

Finally put $L = \log(\tau/\gamma)$. The complete recursive upper envelope is

$$H(\alpha) = \frac{\alpha}{\gamma}\left(u_+ - u_- L + \frac{u_+}{2}L^2\right) - A_3^{\mathrm{low}}(\alpha) - T(\alpha). \tag{14}$$

The original linear-sieve envelope is 4α, so the new saving is

$$\Delta = \int_{7/6}^{5/4} \max\{0, 4\alpha - H(\alpha)\}\, d\alpha. \tag{15}$$

4. Integer certificate for the switching integral

Proposition 4.1. *The integral in* (15) *satisfies*

$$\Delta \geq \frac{32303187971}{10^{12}} = 0.032303187971 > \frac{4}{125}.$$

Proof. We describe the exact certificate implemented in the accompanying file `scripts/certify_harman_recursive_tail.cpp`. It uses $N = 1200$ equal α-cells, $B = 300$ cells for (12), and $C = 160$ cells for the tail (13). All quantities are stored as integer multiples of 10^{-12}. Products use signed 128-bit intermediates; lower bounds are rounded down, upper bounds are rounded up, and every narrowing conversion is checked for overflow.

For an α-cell $[\alpha_i, \alpha_{i+1}]$, monotonicity gives the common domain

$$[\gamma(\alpha_i), \tau(\alpha_{i+1})]$$

and the containing domain

$$[\gamma(\alpha_{i+1}), \tau(\alpha_i)].$$

The first three terms in (14) are bounded, in proof-adverse directions, by

$$\begin{aligned}
A^{+}_{0,i} &= u_{+}\frac{\alpha_{i+1}}{\gamma(\alpha_{i+1})}, \\
A^{-}_{1,i} &= u_{-}\frac{\alpha_{i}}{\gamma(\alpha_{i})}R^{-}_{B}, \\
A^{+}_{2,i} &= \frac{u_{+}}{2}\frac{\alpha_{i+1}}{\gamma(\alpha_{i+1})}(R^{+}_{B})^{2},
\end{aligned}$$

where R^{-}_{B} is the right-endpoint lower Riemann sum for $\int d\beta/\beta$ over the common domain, and R^{+}_{B} is the left-endpoint upper sum over the containing domain.

For the three-prime lower term, each ordered β_1, β_2 box produces a finite family of rational β_3 intervals. Some are uniformly Type II throughout the entire α, β_1, β_2 cell. The others come from the 21 rational lower-sieve thresholds between 2.01 and 4. Overlapping intervals take the largest certified lower weight. The β_3 integral is evaluated exactly by

$$\int_{r}^{s}\frac{d\beta_3}{\beta_3^2} = \frac{1}{r} - \frac{1}{s},$$

and $1/(\beta_1\beta_2)$ is replaced by its value at the upper box endpoints. Thus the resulting $A^{-}_{3,i}$ is a lower bound.

For the recursive tail, a β_1 cell lies in the common intersection $[\tau(\alpha_i), a(\alpha_i)]$. Its child β_2 cells cover a containing domain from $\gamma(\alpha_{i+1})$ to the upper endpoint of the β_1 cell. Each child upper bound is the minimum of the proof-adverse linear-sieve bound, a uniformly available Type II bound, and the recursive base-minus-children bound when the two-prime prefix is at most ξ. The same-exponent-cell region is subtracted in full. This deliberately overcounts both distinct primes in one cell and the harmless region $\beta_2 \geq \beta_1$; it cannot make the tail lower bound too large.

It follows pointwise on the cell that

$$\max\{0, 4\alpha - H(\alpha)\} \geq \max\{0, 4\alpha_i - (A^{+}_{0,i} - A^{-}_{1,i} + A^{+}_{2,i} - A^{-}_{3,i} - T^{-}_{i})\}.$$

Only $\alpha < 1.22$ is retained; zero is used elsewhere.

The exact integer output is

```
scale=1000000000000
alpha_steps=1200 beta_steps=300 tail_beta_steps=160
positive_alpha_cells=727
tail_lower_integral=0.018330264130
a3_lower_integral=0.044459636636
saving_lower_units=32303187971
saving_lower=0.032303187971
target=0.032000000000
CERTIFIED=YES
```

The last comparison is the integer inequality $32303187971 > 32000000000$.

The module `HarmanRecursiveCertificate.lean` is a third Lean implementation of the complete fixed-point algorithm. Lean evaluates the canonical grid by `native_decide` in four 192-cell blocks and one zero block. The respective unscaled saving sums are

154113574025565, 151615832402575, 117510700363272, 41925800002266, 0.

Their sum is 465165906793678; downward division by $12N = 14400$ is 32303187971. Small projection modules connect every expensive block computation to its exact natural-number value. Thus the large block equalities use Lean's native evaluator; after those equalities are available, `HarmanRecursiveCertificateCanonical.lean` checks the final natural-number sum, division, rational normalization, and strict endpoint inequalities by kernel elaboration without re-expanding the blocks.

As an implementation-level cross-check, the accompanying Python audit program independently reconstructs every cell using Python arbitrary-precision integers. It does not call or parse the C++ executable and uses an independent sweep-line algorithm for weighted interval maxima. On the canonical grid it reports the same totals. With the optional `-dump-cells` flag, all 1200 canonical cell records agree byte for byte; both dumps have SHA-256

`6e1901ca1dc8b2c32ae6bcc3759a42ea`
`9a3dab8932870eeb3f6e2d42f0ac3175`.

Thus the Lean, fixed-width, and arbitrary-precision implementations all recompute and certify the displayed lower bound independently, with the Lean block computations using `native_decide` as just stated. □

Remark 4.2. The same source compiled with GCC's undefined-behavior sanitizer gives the same result. The finer grids $1800 \times 400 \times 220$ and $2400 \times 500 \times 280$ give the respective lower bounds 0.033508163100 and 0.034213110567. These values support convergence but are not used in the theorem.

Lemma 4.3 (Explicit analytic-margin budget)**.** *Let H_t be the envelope obtained from the displayed branch ledger by using the shifted parameters in* (4) *with margin t on $7/6 < \alpha < 61/50$; outside that retained range set $H_t = H$. Put*

$$\Delta_t := \int_{7/6}^{5/4} \max\{0, 4\alpha - H_t(\alpha)\}\, d\alpha.$$

For $0 \le t \le 10^{-20}$,

$$|\Delta_t - \Delta| \le 10^{11} t. \tag{16}$$

Consequently the fixed choice

$$\nu_0 = 10^{-20} \tag{17}$$

satisfies all interior inequalities and spends less than half of the exact endpoint margin:

$$10^{11}\nu_0 = 10^{-9} < \frac{1}{10000} < \frac{2994511739}{18000000000000}.$$

Proof. Only $7/6 < \alpha < 61/50$ contributes to the certificate. On this interval

$$\gamma \ge \frac{1}{25}, \qquad \tau \ge \frac{7}{50}, \qquad \frac{1}{2} - \beta_1 - \beta_2 \ge \frac{3}{50} \quad (\tau < \beta_1 < a,\ \beta_2 < \beta_1).$$

Hence, for $0 \le t \le 10^{-20}$,

$$\gamma_t > \frac{1}{50}, \qquad \tau_t > \frac{1}{10}, \qquad \frac{1}{2} - \beta_1 - \beta_2 - t > \frac{1}{20}.$$

All integration variables are therefore at least $1/50$. Moreover $|\log(\tau_t/\gamma_t)| < \log 4 < 2$, $0 \le W_3 \le 1$, and $0 \le U_2 \le U_{\mathrm{LS},t} \le 40$. The absolute densities in the three-prime and two-prime integrals are consequently bounded by

$$\frac{61}{50}\, 50^4 < 8 \cdot 10^6, \qquad \frac{61}{50}\, 40\, 50^2 < 2 \cdot 10^5,$$

respectively; the elementary base term in (14) is below 400. These are explicit integrable majorants on a unit box.

Away from a branch boundary, differentiate the rational branches with respect to t. Since $|\Phi'| \le 1$, $|ds_{3,t}/dt| = 1/\beta_3 \le 50$, and

$$\left| \frac{d}{dt} \frac{2}{1/2 - \beta_1 - \beta_2 - t} \right| \le 800,$$

the base, three-prime, and complete two-prime densities vary by at most $10^6 t$, $4 \cdot 10^8 t$, and $5 \cdot 10^8 t$, respectively. The last bound also covers the recursive candidate: differentiating u_+/γ_t costs less than $10^4 t$, and its inner W_3/β_3^2 integral costs less than $2 \cdot 10^5 t$. Taking a maximum or minimum does not enlarge the largest of these Lipschitz constants.

It remains to account for changing branch indicators and integration limits. There are eight hyperplanes from the four subset sums against the two Type II endpoints, two from the pair Type II interval, one from the recursive ξ_t test, and five occurrences of moving γ_t or τ_t integration limits. Thus even counting repeated priority occurrences separately gives fewer than 32 moving affine hyperplanes. Each right-hand side moves by at most $5t$. Because some coefficient of a moving hyperplane is 1, its symmetric difference inside the unit box has measure at most $10t$. Using the $8 \cdot 10^6$ majorant, all moving slabs contribute less than $32 \cdot 10t \cdot 8 \cdot 10^6 < 3 \cdot 10^9 t$. Together with the smooth variation above and the fact that the α-interval has length below 1, this is less than $10^{11} t$. Finally, $r \mapsto \max(0, r)$ is 1-Lipschitz, which proves (16).

For clarity, no separate grid-limit loss is hidden here. The canonical mesh has α-width $1/14400$, three-prime widths at most $1/3000$, tail β_1-width at most $1/2000$, and containing child width at most $9/8000$. Proposition 4.1 assigns proof-adverse values to every one of these boxes, including boxes meeting Type II or priority boundaries. Thus the locked value 32303187971 is already the post-discard lower bound; the only subsequent loss is the margin comparison (16).

The affine requirements are immediate from $4\nu_0 < 1/25 \le \gamma$ and $5\nu_0 < 1/25 \le \sigma - a$. The last displayed rational comparison is also checked in the finite Lean audit. Reserve the other half of the endpoint margin and then take x large enough to absorb the power-saving and $o(1)$ terms. □

5. An independent lower bound for the old subtractive term

The original decomposition retains

$$F_6 = \int_{7/6}^{5/4} \alpha \int_{\alpha-1}^{\sigma} \omega\left(\frac{\alpha - \beta}{\beta}\right) \frac{d\beta}{\beta^2}\, d\alpha. \tag{18}$$

We need a stronger bound than the conservative 0.035631 printed in the original paper, but it can be obtained without numerical Buchstab integration.

Lemma 5.1. *One has*

$$F_6 > \frac{149403}{2500000} = 0.0597612.$$

Proof. Set $v = \alpha/\beta - 1$. The inner integral in (18) becomes

$$\int_{(4\alpha-2)/(2-\alpha)}^{1/(\alpha-1)} \omega(v)\, dv.$$

Throughout $7/6 \le \alpha \le 5/4$, this interval lies in $[3.2, 6]$. Thus the lower bound in (9) applies and

$$F_6 \ge 0.5607 \int_{7/6}^{5/4} w(\alpha)\, d\alpha, \qquad w(\alpha) = \frac{1}{\alpha - 1} + 4 - \frac{6}{2 - \alpha}.$$

Moreover,

$$w''(\alpha) = \frac{2}{(\alpha-1)^3} - \frac{12}{(2-\alpha)^3} > 0,$$

since the first term is at least 128 and the second at most 768/27. Therefore the composite midpoint sum is a lower bound. Ten equal panels give, by exact rational arithmetic,

$$\begin{aligned} 0.5607 \int_{7/6}^{5/4} w(\alpha)\, d\alpha &\ge \frac{5607}{10000} \frac{1}{120} \sum_{j=0}^{9} w\left(\frac{7}{6} + \frac{2j+1}{240}\right) \\ &= 0.0597612998251544\ldots > 0.0597612. \end{aligned}$$

□

6. The endpoint budget and proof of the theorem

Merikoski's published strict upper bounds for the other terms are

$$F_1 < 0.0287, \qquad F_2 < 0.08622, \qquad F_3 < 0.03107,$$
$$F_4 < 0.00011, \qquad F_5 = \frac{29}{72}.$$

Combining these with Lemma 5.1 and the complete lower bound in Proposition 4.1, the certified zero-margin contribution through exponent 5/4 is strictly smaller than

$$\begin{aligned} C = &\frac{1}{6} + \frac{287}{10000} + \frac{8622}{100000} + \frac{3107}{100000} + \frac{11}{100000} + \frac{29}{72} \\ &- \frac{149403}{2500000} - \frac{32303187971}{10^{12}} \\ = &\frac{5611320508261}{9000000000000}. \end{aligned} \tag{19}$$

Above 5/4, the unchanged dimension-one linear sieve contributes through a block exponent ϖ

$$4\int_{5/4}^{\varpi} \alpha\, d\alpha = 2\left(\varpi^2 - \left(\frac{5}{4}\right)^2\right). \tag{20}$$

Choose

$$\varpi_* := \frac{13231}{10000} = 1.3231.$$

Exact rational arithmetic gives

$$C + 2\left(\varpi_*^2 - \left(\frac{5}{4}\right)^2\right) = \frac{8997005488261}{9000000000000} < 1, \tag{21}$$

$$1 - \frac{8997005488261}{9000000000000} = \frac{2994511739}{9000000000000}. \tag{22}$$

At the fixed analytic margin ν_0, Lemma 4.3 increases the upper bound by at most 10^{-9}, which is less than half the last displayed margin. Choose x large enough that all power-saving, fundamental-lemma, and smooth-partition errors together use less than the remaining half. The resulting normalized total is still strictly below 1.

Let P_x denote the greatest prime factor among the values n^2+1 selected by the smooth index block. The Chebyshev–Hooley identity used in [6] is

$$\sum_{x<p\le P_x} |\mathcal{A}_p| \log p = X_b \log x + O(x). \tag{23}$$

If $P_x \le x^{\varpi_*}$, summing the smooth modulus blocks and applying (19), (20), and Lemma 4.3 gives a strict upper bound smaller than $X_b \log x$, contradicting (23) for all sufficiently large x. Hence every sufficiently large index block $[x, 2x]$ contains an n such that

$$P^+(n^2+1) > x^{1.3231}. \tag{24}$$

Finally put $a = 1323/1000 = 1.323$ and $b = 13231/10000 = 1.3231$. Then $b-a = 1/10000$ and $a/(b-a) = 13230$. For $x > 2^{13230}$ and $n \le 2x$,

$$n^a \le (2x)^a = 2^a x^a < x^{b-a} x^a = x^b.$$

Together with (24), this proves Theorem 1.1.

7. Formal verification and scope

The focused Lean project contains the finite weighted Buchstab identity, its disjoint first-bad-index iteration, the sign pattern $A_0 - A_1 + A_2 - A_3$, the recursive node upper/lower rules, and the subtractive-tail inequality used in Section 3. Its rational-envelope module proves the elementary lower envelope used in the finite certificate. The recursive-certificate module and its four block modules independently recompute the exact outward-rounded certificate; the value and canonical modules aggregate the block totals while keeping peak elaboration memory bounded. The F_6 module proves positivity of the displayed second derivative, convexity, the ten-panel midpoint integral lower bound, and the exact rational comparison with 149403/2500000; applying this width certificate to F_6 still uses the quoted Buchstab lower bound. The endpoint module checks the exact rational comparisons, including $1323/1000 < 13231/10000$ and the strict normalized budget. The real-power module proves the final implication $x > 2^{13230}$ and $0 \le n \le 2x$ imply $n^{1323/1000} < x^{13231/10000}$. The focused project builds with Lean 4 and Mathlib v4.22.0.

Small audit modules check affine identities for the ν margins, an explicit rational $\nu = 10^{-20}$ that satisfies those identities on $7/6 < \alpha < 1.22$, a half-integer cutoff, an abstract count of representations, one-sided box geometry, algebraic normalization of α/γ, $t+1$, Φ, and U_{LS}, the $s<1$ cutoff identity $(2/s)/((1-\omega)\log D) = 2/\log D$, the exact half-margin split 2994511739/18000000000000, the comparison $10^{11}\nu <$ 2994511739/18000000000000, and the finite Perron budget arithmetic under stated hypotheses $\log((V-1/2)/U) \ge 2/(4U+1)$ and $X^B \cdot 2 \cdot X^{C-A'} \le X^{-A}$. These modules construct neither the actual Perron localization nor Rosser coefficients varying with the prefix, and they do not prove any of the six analytic transfer gates as a whole.

The formal development treats Grimmelt–Merikoski Corollaries 7.1 and 7.2 and the standard upper and lower dimension-one linear-sieve theorem as external analytic inputs. Sections 2 and 3 verify their hypotheses and prove the finite Perron/Mellin localization on paper; these analytic arguments are not Lean theorems.

The finite Buchstab decomposition, the integer computation, its primitive outward-rounding rules, the elementary F_6 convex midpoint certificate, the exact rational endpoint arithmetic, and the real-power dyadic conversion are checked in Lean. The expensive $1200 \times 300 \times 160$ block equalities are discharged by `native_decide`; the canonical sum, division, and endpoint comparisons are then checked without native recomputation. Thus the native compiler is a trust boundary only for block values independently cross-checked by C++ and Python; downstream arithmetic and the finite lemmas on signs, widths, clipping, and endpoint directions are kernel checked. None of these facts asserts that the integer algorithm equals the analytic $H(\alpha)$ integral. No floating-point or proprietary computer-algebra calculation is an input to the finite certificate.

Remark 7.1 (AI-assisted proof development). The author discloses that GPT-5.6 Sol was used as a reasoning and coding assistant in developing the core recursive Buchstab switching argument, the proof-direction and branch ledger, the Lean certificate implementation, the exact-integer audit scripts, and the manuscript exposition. The author reviewed the final mathematical statements and takes responsibility for the submitted content. GPT-5.6 Sol is an AI-assisted development tool, not an author or an independent verifier. The Lean project checks only the finite combinatorial, integer, rounding, elementary F_6, rational endpoint, and real-power conversion components; the cited analytic Type-I/II and linear-sieve theorems remain external inputs. The finite Mellin–Perron transfer is proved in the paper but not formalized in Lean.

Remark 7.2 (Supplementary material). The accompanying source archive contains the Lean project and locked toolchain, the fixed-width C++ certificate `certify_harman_recursive_tail.cpp`, and the independent arbitrary-precision Python audit `audit_harman_recursive_tail.py`. The command line `1200 300 160` selects the canonical grid in both external implementations; each prints `saving_lower` = `0.032303187971` and `CERTIFIED=YES`. No precomputed binary data are needed. The immutable source archive is `https://doi.org/10.5281/zenodo.22140874`. The separate script `audit_harman_recursive_branch_coverage.py` exhaustively checks the analytic branch gates summarized in Remark 3.6. A supplied build script packages the paper, PDF, locked Lean project, audits, and analytic ledger into one checksummed review archive. The archive also includes a source-version ledger with theorem locations and SHA-256 hashes for the external PDFs; the external PDFs themselves are not redistributed. The public verification repository is `https://github.com/CGandGameEngineLearner/landau-fourth-problem-1.323`.

Remark 7.3 (Analytic review boundary). The finite cross-condition localization and prefix-uniform sieve are proved in Lemma 3.1 and Proposition 3.2, but they are not formalized in Lean. Together with the quoted Type I/II and linear-sieve theorems, these are the parts that require conventional analytic-number-theory refereeing.

Remark 7.4 (Distance from Landau's problem). Theorem 1.1 implies infinitely often a factorization

$$n^2 + 1 = pm, \qquad p > n^{1.323}, \qquad m < n^{0.677+o(1)},$$

but it neither forces $m = 1$ nor overcomes the parity barrier. It is a new greatest-prime-factor weak form of Landau's fourth problem, not a proof of the prime-values conjecture.

Appendix A. Details for the prefix-uniform linear sieve

This appendix records the analytic bookkeeping used in Proposition 3.2; the main text contains the proof route and the limitation of the argument.

A.1. **One localized prefix.** Fix one exponent cell and an ordered prefix $\boldsymbol{q} = (q_1, \dots, q_j)$, $j \le 3$, and put $u = q_1 \cdots q_j$. The prefix partition is kept outside the sifted cofactor sum; write its nonnegative weight as $c(\boldsymbol{q})$. No prefix-dependent cofactor cutoff is needed. Define

$$a_u(m) := |\mathcal{A}_{um}| \psi_P(um) \log(um), \qquad |\mathcal{A}_{u,d}| := \sum_{d|m} a_u(m),$$

and, with the same weight, the model sums

$$|\mathcal{B}_{u,d}| := X_b \sum_{d|m} \frac{\rho(um)}{um} \psi_P(um) \log(um), \qquad X_u := |\mathcal{B}_{u,1}|.$$

Write

$$\begin{aligned} \Delta(k) &:= \sum_{\ell^2+1 \equiv 0 \pmod{k}} b(\ell) - X_b \frac{\rho(k)}{k}, \\ r_u(d) &:= |\mathcal{A}_{u,d}| - |\mathcal{B}_{u,d}|, \\ r_P(e) &:= \sum_{e|k} \psi_P(k) \log k \, \Delta(k). \end{aligned} \tag{25}$$

A Rosser modulus $d \mid P(z(\boldsymbol{q}))$ has all prime factors below the least prefix prime, so $(u, d) = 1$. The changes of variables $k = um$ and $e = ud$ are then exact and give the important identity

$$r_u(d) = r_P(ud). \tag{26}$$

Thus the discrepancy no longer remembers which ordered prefix represents u.

For completeness, the model evaluation is uniform in the prefix. Terms with $(m, u) > 1$ have an extra prime divisor at least x^{γ_ν} and contribute $O(x^{-\gamma_\nu + o(1)} X_u)$, uniformly because $j \le 3$. On the remaining terms $\rho(um) = \rho(u)\rho(m)$. Partial summation and the Euler-product calculation of [6, Lemma 8] and [3, Theorem 11.12] therefore give the same dimension-one local factors as in the unprefixed sequence. In particular,

$$V_u(z) = \prod_{\substack{p < z(\boldsymbol{q}) \\ p \nmid u}} (1 - g(p)) = \prod_{p < z(\boldsymbol{q})} (1 - g(p)).$$

Indeed, after writing $q_i = Q_i v_i$ and $k = Pw$, every smooth factor is drawn from one fixed compactly supported family in the normalized variables v_i, w; its rth derivative is $O_r(1)$, independently of the exact prefix and of x. For $p < z = q_j$ one has $p \nmid u$, so the local density is exactly $g_u(p) = g(p)$. Consequently the dimension-one product condition used in Chapters 11–12,

$$\prod_{w \le p < z} (1 - g_u(p))^{-1} \le \frac{\log z}{\log w} \left(1 + O\left(\frac{1}{\log w}\right)\right),$$

has the same implied constant as the unprefixed product. The fundamental lemma constants depend only on this product bound and the fixed derivative seminorms, not on the value of the least prefix prime inside its dyadic cell. This proves the required uniformity when $z = q_j$.

More explicitly, summing over the ordered prefix cells gives

$$\sum_{\boldsymbol{q}} c(\boldsymbol{q}) \sum_{i \le j} \sum_{q_i \mid m} X_b \frac{\rho(um)}{um} \psi_P(um) \log(um) \ll x^{-\gamma_\nu + o(1)} \mathfrak{M}(P),$$

while the coprime part has the Chapter 11 remainder $o(\mathfrak{M}(P))$, uniformly on a fixed cell. Thus the aggregate model-evaluation error is $o(\mathfrak{M}(P))$. This is the same calculation used in Merikoski's proof of Lemma 8; the only new factor $\rho(u)/u$ is outside the cofactor sieve.

Let $\lambda^{\pm}_{d,\boldsymbol{q}}$ be the level-D Rosser weights. They are supported on squarefree $d \mid P(z(\boldsymbol{q}))$, $d < D$, and satisfy $|\lambda^{\pm}_{d,\boldsymbol{q}}| \le 1$. The Rosser inequalities and the preceding model evaluation give, with $E_u \ge 0$ and $\sum_{\boldsymbol{q}} c(\boldsymbol{q}) E_u = o(\mathfrak{M}(P))$, [3, Chapter 12, (12.12)–(12.14), pp. 235–236]

$$S(\mathcal{A}(P)_u, z(\boldsymbol{q})) \le X_u V_u(z)\{F(s_{\boldsymbol{q}}) + O((\log D)^{-1/6})\} + R^{+}_{\boldsymbol{q}} + E_u, \tag{27}$$

$$S(\mathcal{A}(P)_u, z(\boldsymbol{q})) \ge X_u V_u(z)\{f(s_{\boldsymbol{q}}) - O((\log D)^{-1/6})\} + R^{-}_{\boldsymbol{q}} - E_u. \tag{28}$$

where $s_{\boldsymbol{q}} = \log D / \log z(\boldsymbol{q})$ and

$$R^{\pm}_{\boldsymbol{q}} = \sum_{\substack{d \mid P(z(\boldsymbol{q})) \\ d < D}} \lambda^{\pm}_{d,\boldsymbol{q}} r_P(ud).$$

The signs of these remainders are retained: no absolute value is taken before the prefix sum is collected. The upper formula is used directly for $s_{\boldsymbol{q}} \ge 1$; the lower formula is used for $s_{\boldsymbol{q}} \ge 2$, and zero is used below 2.

A.2. **Collecting the prefix sum.** For a cell with $u \le x^{s_0}$, take

$$D = x^{1/2 - s_0 - \nu}, \qquad ud \le x^{1/2-\nu}.$$

Let $c(\boldsymbol{q}) \ge 0$ denote the smooth dyadic/exponent partition of the ordered prefix; its size and all partial-summation losses are $O((\log x)^B)$. Multiply (27) or (28) by $c(\boldsymbol{q})$, sum first, and only then take an absolute value. Using (26) and collecting $e = ud$ gives the exact identity

$$\mathcal{R}^{\pm} := \sum_{\boldsymbol{q}} c(\boldsymbol{q}) R^{\pm}_{\boldsymbol{q}} = \sum_{e \le x^{1/2-\nu}} \Lambda^{\pm}_e r_P(e), \qquad \Lambda^{\pm}_e := \sum_{\substack{\boldsymbol{q}, d\colon ud = e \\ d \mid P(z(\boldsymbol{q})),\ d < D}} c(\boldsymbol{q}) \lambda^{\pm}_{d,\boldsymbol{q}}. \tag{29}$$

If an auxiliary smooth product cutoff is introduced, Mellin inversion first writes it as a polylogarithmic-L^1 superposition of the same identity with a common smooth function of k/P; Corollary 7.1 is uniform for each such function. Hence it creates only the displayed logarithmic loss, not a prefix-dependent discrepancy inside the k-sum.

For fixed e, the at most three ordered prefix primes divide e, and $d = e/u$ is determined. Hence fixed divisor bounds for $c(\boldsymbol{q})$ and $|\lambda^{\pm}_{d,\boldsymbol{q}}| \le 1$ imply

$$|\Lambda^{\pm}_e| \ll (\log x)^B \tau_K(e) \tag{30}$$

with fixed B, K. Put

$$\delta_0 = (1 - 2\theta)\nu - \varepsilon_{\mathrm{GM}}/4 > 0.$$

Split $[P, 4P]$ into a fixed number of smooth dyadic blocks and remove $\log k$ by partial summation. Corollary 7.1 of [2], with $a = h = 1$, $\eta = \nu$, and $P \le x^2$, has inside its modulus-e absolute value exactly

$$\sum_{k \equiv 0 \pmod e} \psi_1(k/K) \left\{ \sum_{\ell^2+1 \equiv 0 \pmod k} \psi_2(\ell/x) - \frac{\rho(k)}{k} x\widehat{\psi}_2(0) \right\}.$$

Rescaling b supplies ψ_2, the fixed decomposition of ψ_P supplies ψ_1, and partial summation supplies $\log k$; hence this is precisely $r_P(e)$ in (25). Corollary 7.1 therefore gives

$$\sum_{e \le x^{1/2-\nu}} |r_P(e)| \ll x^{1-\delta_0} (\log x)^{B_0}.$$

Choosing $\kappa < 2\delta_0$ in $\tau_K(e) \ll e^\kappa$, and absorbing the dyadic logarithms, yields

$$\begin{aligned} |\mathcal{R}^\pm| &\le \sum_{e \le x^{1/2-\nu}} |\Lambda_e^\pm| \, |r_P(e)| \\ &\ll (\log x)^B x^{\kappa/2} \sum_{e \le x^{1/2-\nu}} |r_P(e)| \\ &\ll (\log x)^{B+B_0} x^{1-\delta_0+\kappa/2} \ll x^{1-\delta}. \end{aligned} \tag{31}$$

for some $\delta = \delta(\nu) > 0$. Thus only the total multiple sum, not each prefix separately, has the required power-saving error.

A.3. **The two sieve branches.** For the A_3 lower branch, $u = q_1q_2q_3$, $z = q_3$, and

$$\begin{aligned} s_{3,\nu} &= \frac{1/2 - \beta_1 - \beta_2 - \beta_3 - \nu}{\beta_3}, \\ \widehat{s}_{3,\nu} &= \frac{1/2 - s_0 - \nu}{\beta_3} + O(1/\log x) \le s_{3,\nu} + O(1/\log x). \end{aligned}$$

The cell upper endpoint therefore gives a lower sieve weight. Replace $e^{-\gamma_E} f(\widehat{s}_{3,\nu})$ by $\Phi(\widehat{s}_{3,\nu})$; if $\widehat{s}_{3,\nu} \le 2$, use zero. At the endpoint $\alpha = 7/6$,

$$3\tau_\nu = \frac{1}{2} - \frac{9}{2}\nu, \qquad \frac{1}{2} - s_0 - \nu > \frac{7}{2}\nu,$$

so D remains a positive power of x. This proves the three-prime remainder is admissible at the fixed margin $\nu = \nu_0$.

For the two-prime upper branch, $u = q_1q_2$, $z = q_2$, and

$$\widehat{s}_{2,\nu} = \frac{1/2 - s_0 - \nu}{\beta_2} + O(1/\log x) > 0.$$

If $1 \le \widehat{s}_{2,\nu} \le 3$, then $e^{-\gamma_E} F(\widehat{s}_{2,\nu}) = 2/\widehat{s}_{2,\nu}$; if $\widehat{s}_{2,\nu} \ge 3$, use $2/(3\beta_2)$. When $0 < \widehat{s}_{2,\nu} < 1$, put $z_* = D^{1-\omega} < D < q_2$. Positivity gives

$$S(\mathcal{A}(P)_{q_1q_2}, q_2) \le S(\mathcal{A}(P)_{q_1q_2}, z_*),$$

whose sieve parameter is $1/(1-\omega) > 1$. For fixed $\omega > 0$ the upper-sieve formula applies, and

$$\frac{e^{-\gamma_E} F(1/(1-\omega))}{\log z_*} = \frac{2}{\log D}.$$

After $x \to \infty$, letting $\omega \to 0^+$ therefore gives the first term in

$$U_{\mathrm{LS},\nu}(\beta_1, \beta_2) = \max\left\{\frac{2}{1/2 - \beta_1 - \beta_2 - \nu}, \frac{2}{3\beta_2}\right\}.$$

The finite mesh uses the proof-adverse endpoints; boxes without positive level contribute no certified tail saving.

Finally, for $z(\boldsymbol{q}) \in [Z, 2Z]$, $Z = x^{\zeta}$,

$$\frac{\log D}{\log z(\boldsymbol{q})} = \frac{1/2 - s_0 - \nu}{\zeta} + O(1/\log x).$$

The functions F, f are continuous at the transition points, and the certificate uses the lower value for the lower branch and the upper value for the upper branch. Hence freezing the multiplier on a cell changes the total model sum by $o(\mathfrak{M}(P))$. Merikoski's Lemma 7 and the prime number theorem then give, on an A_3 cell,

$$(1 + o(1))\mathfrak{M}(P)\,\alpha \int_{\text{cell}} e^{-\gamma_E} f(s_{3,\nu}) \frac{d\beta_1 d\beta_2 d\beta_3}{\beta_1\beta_2\beta_3^2}.$$

Since $\Phi(s) \le e^{-\gamma_E} f(s)$, summing the cells gives (12). For the two-prime upper branch the corresponding coefficient relative to $d\beta_1 d\beta_2/(\beta_1\beta_2)$ is $e^{-\gamma_E} F(s_{2,\nu})/\beta_2$. It equals $2/(1/2 - \beta_1 - \beta_2 - \nu)$ for $1 \le s_{2,\nu} \le 3$, is at most $2/(3\beta_2)$ for $s_{2,\nu} \ge 3$, and has the first bound after the cutoff reduction when $0 < s_{2,\nu} < 1$. This is precisely $U_{\mathrm{LS},\nu}$, and summing gives its term in (13). All fundamental-lemma and Type-I estimates used here are the cited external analytic inputs; the prefix collection itself is the exact calculation (29).

References

[1] N. A. Carella, *Largest prime factors of polynomials*, arXiv:2308.10075v7, 2025.

[2] L. Grimmelt and J. Merikoski, *On the greatest prime factor and uniform equidistribution of quadratic polynomials*, arXiv:2505.00493v2, 2025.

[3] J. Friedlander and H. Iwaniec, *Opera de Cribro*, American Mathematical Society Colloquium Publications, vol. 57, American Mathematical Society, Providence, 2010.

[4] H. Iwaniec, *Almost-primes represented by quadratic polynomials*, Invent. Math. **47** (1978), 171–188, doi:10.1007/BF01578070.

[5] R. Li, *On the largest prime factor of quadratic polynomials*, arXiv:2406.07575v2, 2025.

[6] J. Merikoski, *On the largest prime factor of $n^2 + 1$*, J. Eur. Math. Soc. (JEMS) **25** (2023), 1253–1284, doi:10.4171/JEMS/1216.

[7] A. Pascadi, *Large sieve inequalities for exceptional Maass forms and the greatest prime factor of $n^2 + 1$*, Forum Math. Pi **14** (2026), e8, 1–54, doi:10.1017/fmp.2026.10025.

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